Agile Principles and Mindset

Lecture 6

ASTANA IT UNIVERSITY

Why Use Agile?

Different types of projects require different methods.

Ideal for knowledge worker projects that occur in fast-moving, complex environment.

ASTANA IT UNIVERSITY

Knowledge Worker Projects Are Different

Knowledge worker: Person with subject matter expertise who communicates their knowledge and takes part in analysis and/or development efforts

Examples: Engineers, teachers, scientists, doctors, etc.

Characteristics of Industrial Work	Characteristics of Knowledge Work
Work is visible	Work is invisible
Work is stable	Work is changing
More structure with fewer decisions	Less structure with more decisions
Define the task	Understand the task
Strict standards	Continuous innovation
Focus on quantity	Focus on quality
Focus on the right answers	Focus on the right questions

ASTANA IT UNIVERSITY

Defined Versus Empirical Processes

Defined process: Defines the constituent steps in advance, as construction projects

Empirical process: Iterative and incremental, as with new or uncertain projects

ASTANA IT UNIVERSITY

The Agile Mindset

General characteristics of the agile mindset as written in the Declaration of Interdependence:

1. Increase return on investment
2. Deliver reliable results
3. Expect uncertainty
4. Unleash creativity and innovation
5. Boost performance
6. Improve effectiveness and reliability

ASTANA IT UNIVERSITY

Being Agile Versus Doing Agile

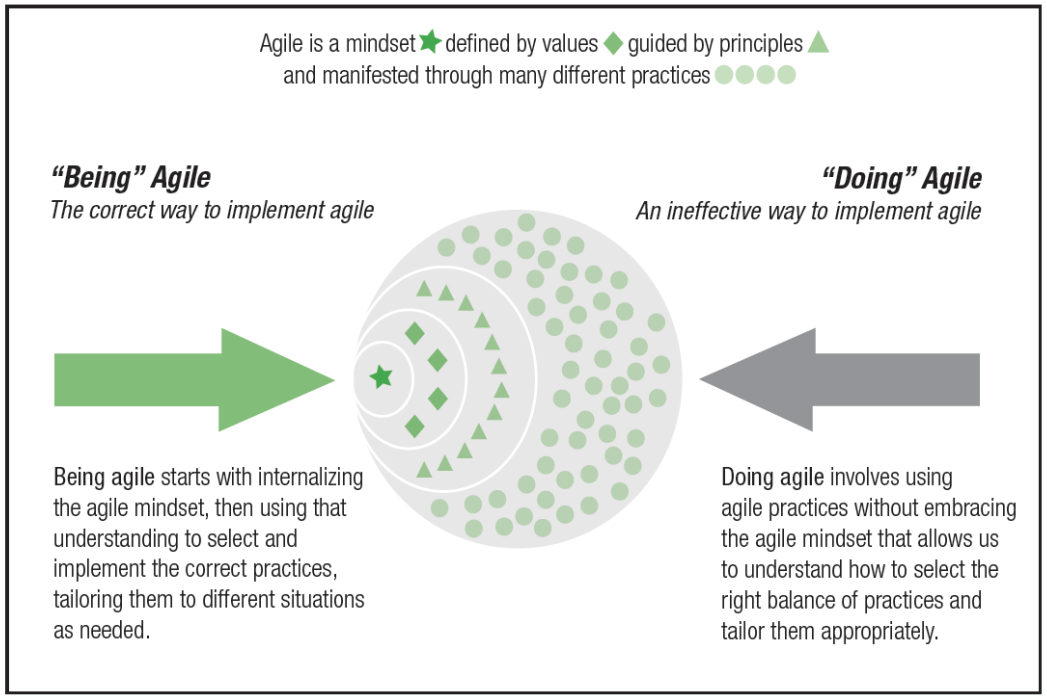


Image copyright © Ahmed Sidky, Santeon Group, www.santeon.com

ASTANA IT UNIVERSITY

Personal, Team, and Organizational Agility

The more people embrace and act upon agile principles, the more effective they will be.

Organizations change through influence exerted by individuals.

Process of creating organizational change:

Think: individually learning and internalizing agile principles

Do: doing the practice of agile

Encourage Others: encourage others to become agile

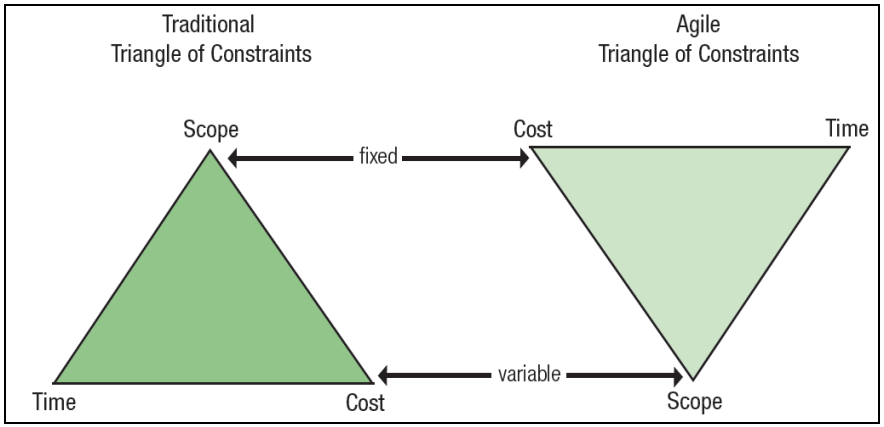
ASTANA IT UNIVERSITY

The Agile Triangle

Agile teams allow scope to vary within the fixed parameters of cost and time.

Knowledge worker projects are characterized by experimentation and uncertainty.

Inverted Triangle Model



ASTANA IT UNIVERSITY

Agile Manifesto

Created by software and methodology experts at the forefront of emerging agile methods to highlight the agile mindset and the values and principles it is based on.



ASTANA IT UNIVERSITY

The 4 Values of the Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions over processes and tools

Working software over comprehensive documentation

Customer collaboration over contract negotiation

Responding to change over following a plan

While there is value in the items on the right, we value the items on the left more.

ASTANA IT UNIVERSITY

The 4 Values of the Agile Manifesto (Continued)

Individuals and interactions over processes and tools

- Projects are undertaken by people, not tools
- Problems get solved by people, not processes

Working software over comprehensive documentation

- Emphasizes the need to deliver
- Focuses on the business value, rather than paperwork

ASTANA IT UNIVERSITY

The 4 Values of the Agile Manifesto (Continued)

Customer collaboration over contract negotiation

- Be flexible and accommodating
- Work with the customer toward a shared definition of “done”

Responding to change over following a plan

- Acknowledge that initial plans were made when the least was known about the project
- Effort should be spent responding to inevitable changes

ASTANA IT UNIVERSITY

Agile Manifesto:12 Principles

We follow these principles:

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

4. Business people and developers must work together daily throughout the project.

ASTANA IT UNIVERSITY

Agile Manifesto:12 Principles (Continued)

5. Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.

6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

7. Working software is the primary measure of progress.

8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

ASTANA IT UNIVERSITY

Agile Manifesto:12 Principles (Continued)

9. Continuous attention to technical excellence and good design enhances agility.

10. Simplicity—the art of maximizing the amount of work not done—is essential.

11. The best architectures, requirements, and designs emerge from self-organizing teams.

12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

ASTANA IT UNIVERSITY

Agile Methodologies

Over 12 actively used agile methodologies

Common approaches include:

Scrum

Extreme Programming (XP)

Lean product development

Kanban

Feature-Driven Development (FDD)

Dynamic Systems Development Method (DSDM)

Crystal family of methods

Most widely used are Scrum and XP

ASTANA IT UNIVERSITY

Scrum

Lightweight, easy to understand model

Set of practices, roles, events, artifacts, and rules designed to guide the team in executing the project

Based on three pillars:

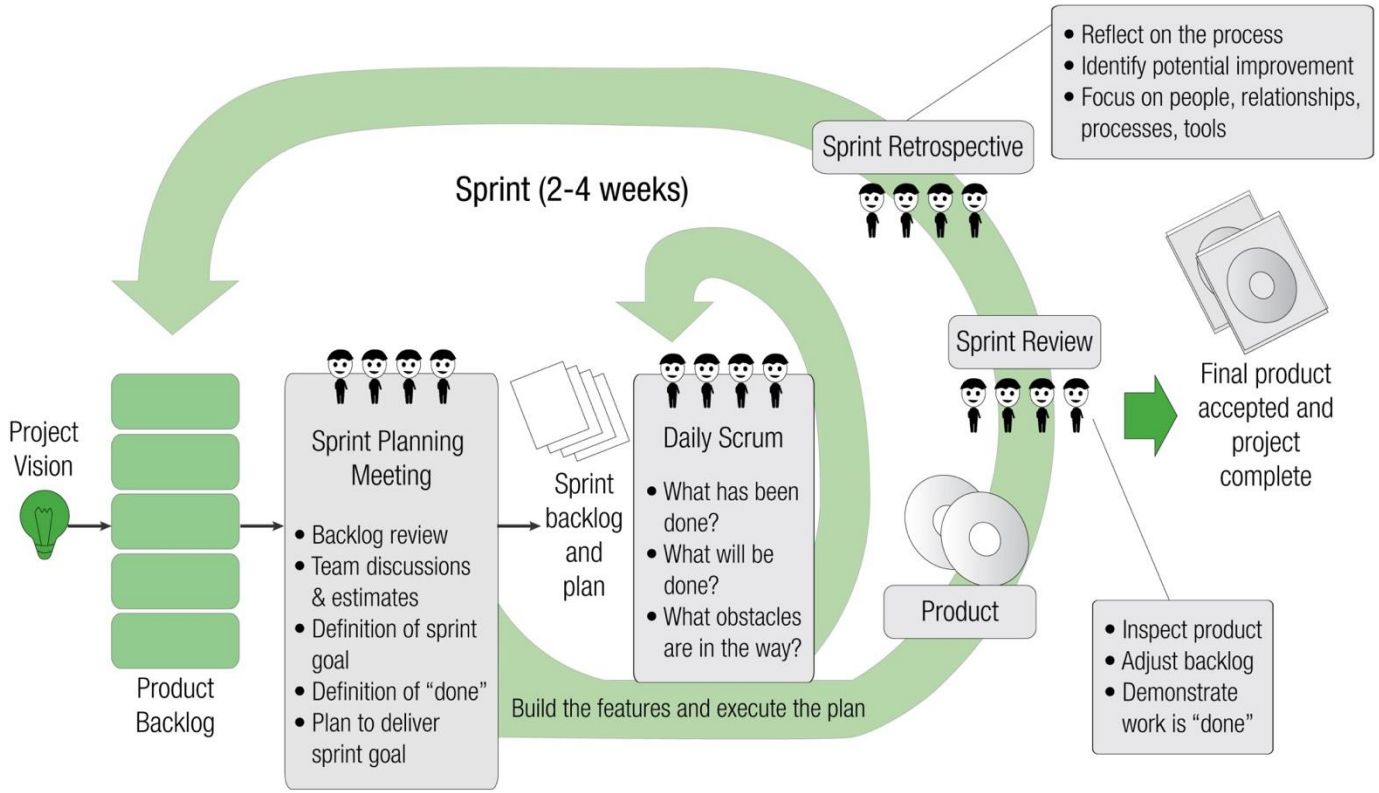
Transparency: Give visibility to those responsible for the outcome of the project

Inspection: Conduct timely checks on how well a project is progressing toward goals

Adaptation: Adjust a process to minimize further issues if inspection shows a problem

ASTANA IT UNIVERSITY

Scrum Process



ASTANA IT UNIVERSITY

Sprints

- Timeboxed (time-limited) iteration
  - One month of lesson
  - Team builds a potentially releasable product

- One or two-week long mini-project
  - No changes are made that would affect the sprint goal

- Scrum “sprint” = Generic agile “iteration”

ASTANA IT UNIVERSITY

Scrum Team Roles

- Development team: Self-organizing group of people who
  build the product increments

- Product owner: Manages the product backlog (list of work
  to be done) with the goal of maximizing the product value

- ScrumMaster: Servant leader ensures that the Scrum methodology is understood and used effectively

ASTANA IT UNIVERSITY

Scrum Activities (Events, Ceremonies)

- Backlog refinement: Meeting to discuss and update the backlog items
- Sprint planning meeting: Used to determine what will be delivered in the sprint and how the work will be achieved
- Daily scrum: Daily 15-minute timeboxed meeting of the development team; assess progress toward the sprint goal
- Sprint review: Meeting at end of the sprint; inspect the increment or product and update the backlog
- Sprint retrospective: Meeting at end of the sprint; team reflects on their processes and looks for improvements

ASTANA IT UNIVERSITY

Scrum Artifacts

- Product increment: Development team members build an increment of the solution, demo the increment, and get feedback

- Product backlog: Dynamic and evolving prioritized list of all the work to be done that serves as the single source of requirements

- Sprint backlog: Subset of items from the product backlog that have been selected as the goal of a sprint

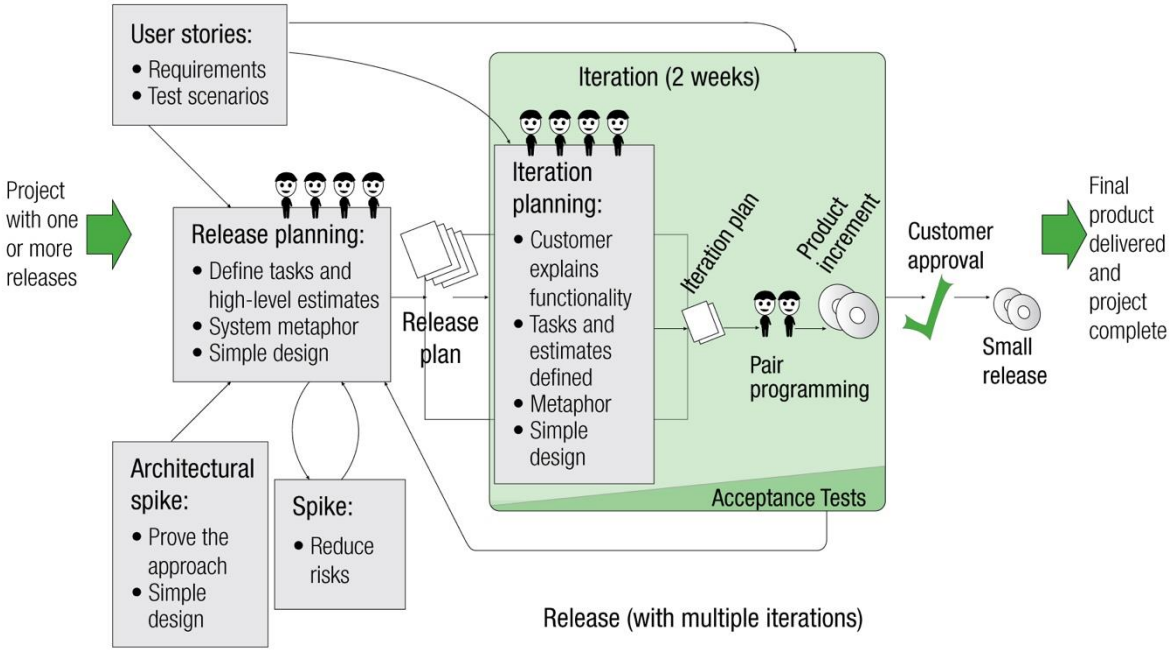
ASTANA IT UNIVERSITY

Extreme Programming (XP)

- Approach focused on software development best practices
- Embraces the following core values:
- Simplicity: Reduce complexity, extra features, and waste
- Communication: Make sure all team members know what is expected of them and what others are working on
- Feedback: Get impressions of suitability early
- Courage: Have courage to allow work to be entirely visible
- Respect: Have respect for others when working as a team; everyone is accountable for the success or failure of the project

ASTANA IT UNIVERSITY

XP Life Cycle



XP Team Roles

- Coach: Mentor and facilitator who helps team stay effective and on track by reinforcing communication with and across teams
- Customer: Business representative who provides project requirements and priorities, and confirms final product works as intended
- Programmers: Developers who build the product by writing and implementing code for requested user stories
- Testers: Provide quality assurance and help customer define and write acceptance tests

ASTANA IT UNIVERSITY

XP Core Practices

Whole team: Contributors sit together in the same location, as members of the same team

Planning games:

- Release planning—A release is a push of new functionality to the production user; project typically has one or more releases
- Iteration planning—An iteration is a short development cycle within a release; in Scrum, these are called “sprints”

Small releases: Used to demonstrate progress, increase visibility, and rapidly deploy working software to end user

ASTANA IT UNIVERSITY

XP Core Practices (Continued)

- Customer tests: Customer describes one or more tests that will show that the software is working as intended
- Collective code ownership: Any pair of developers can improve or amend any code
- Code standards: Teams follow consistent coding standard
- Sustainable pace: Highest level of productivity is achieved by a team operating at a sustainable pace
- Metaphor: Use metaphors and similes to explain designs and create a shared vision that all stakeholders can understand

ASTANA IT UNIVERSITY

XP Core Practices (Continued)

- Continuous integration: Make sure code compiles and works together
- Test-driven development: Team writes the tests before developing the code
- Refactoring: Improve design of existing code without altering external behavior or adding new functionality
- Simple design: Simple, but adequate design allows teams to develop code quickly and adapt as necessary
- Pair programming: Production code is written by two developers working as a pair

ASTANA IT UNIVERSITY

Lean Software Development

- Derived from lean manufacturing approaches
- Closely aligned with agile, but not an agile methodology
- Core concepts include:
  - Using visual management tools
  - Identifying customer-defined value
  - Building in learning and continuous improvement

ASTANA IT UNIVERSITY

Lean Core Concepts

- Eliminate waste: In order to maximize value, remove delays, unnecessary features, partially done work, etc.
- Empower the team: Respect team members' knowledge of the technical steps required and let them make decisions
- Deliver fast: Maximize the project ROI by quickly producing valuable deliverables
- Optimize the whole: See the system as more than the sum of its parts and how it aligns with the organization

ASTANA IT UNIVERSITY

Lean Core Concepts (Continued)

- Build quality in: Continually assure quality throughout the development process
- Defer decisions: Balance early planning with making decisions and commitments as late as possible
- Amplify learning: Facilitate communications early and often, getting frequent feedback

ASTANA IT UNIVERSITY

The 7 Wastes of Lean

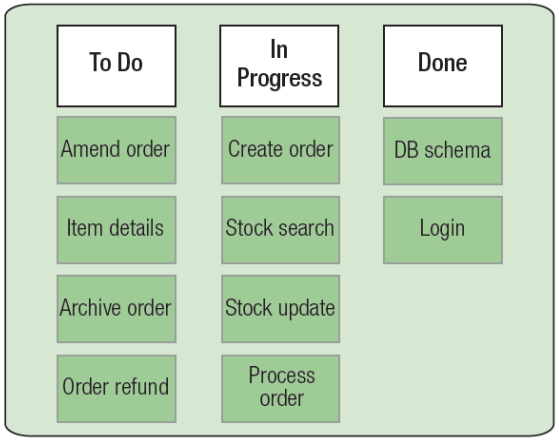
Waste	Description	Example
Partially done work	Work started, but not complete; partially done work can entropy	Code waiting for testing Specs waiting for development
Extra processes	Extra work that does not add value	Unused documentation Unnecessary approvals
Extra features	Features that are not required, or are thought of as “nice-to-haves”	Gold-plating Technology features
Task switching	Multitasking between several different projects when there are context-switching penalties	People assigned to multiple projects
Motion	Effort required to communicate or move information or deliverables from one group to another; if teams are not co-located, effort may be greater	Distributed teams Handoffs
Defects	Defective documents or software that needs correction	Requirements defects Software bugs

ASTANA IT UNIVERSITY

Kanban

- Derived from Toyota's lean production system
- "Kanban" is a Japanese word meaning "signboard"
- Focuses on limiting work in progress (WIP)
- to identify issues and minimize waste

Kanban/task board



ASTANA IT UNIVERSITY

5 Principles of Kanban

1. Visualize the workflow: Visualize the workflow to help organize, optimize, and track it

2. Limit WIP: Restrict amount of work in progress to improve productivity and increase visibility of bottlenecks

3. Manage flow: Track the flow of work through a system to identify issues and measure effectiveness of changes

4. Make process policies explicit: Clearly explain how things work so the team can have open discussions

5. Improve collaboratively: Collectively own and improve processes through scientific measurement and experimentation

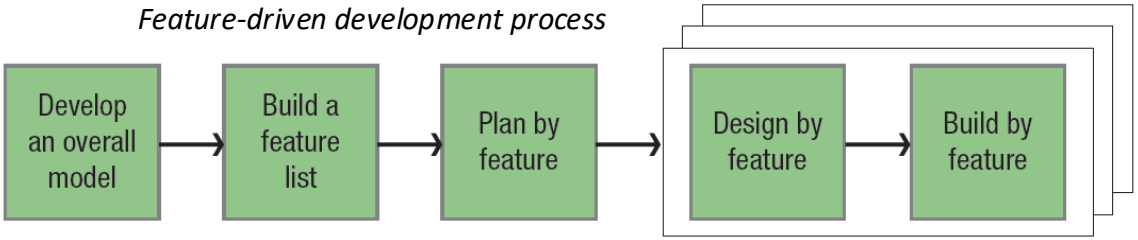
ASTANA IT UNIVERSITY

Feature-Driven Development (FDD)

Recommends a set of good practices, including:

- Domain object modeling
- Developing by feature
- Individual class (code) ownership
- Feature teams
- Inspections
- Configuration management
- Regular builds
- Visibility of progress and results

Feature-driven development process



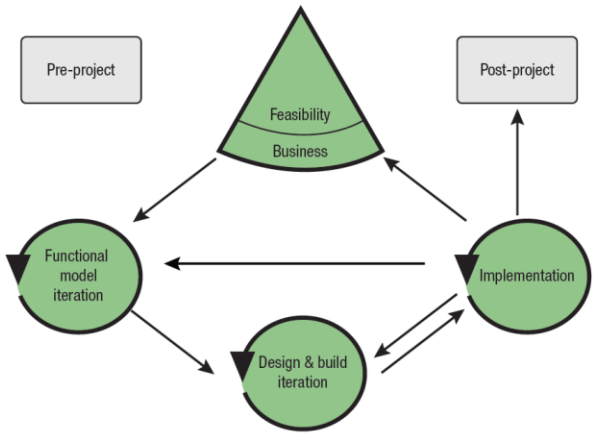
ASTANA IT UNIVERSITY

Dynamic Systems Development Method (DSDM)

Centered on 8 principles, including:

1. Focus on the business need
2. Deliver on time
3. Collaborate
4. Never compromise quality
5. Build incrementally from firm foundations
6. Develop iteratively
7. Communicate continuously and clearly
8. Demonstrate control

DSDM method



ASTANA IT UNIVERSITY

Crystal

Family of situationally specific, customized methodologies designed for different types of projects

Uses factors of criticality and team size to classify projects

Crystal family of methods

Criticality*	Crystal Clear	Crystal Yellow	Crystal Orange	Crystal Red	Crystal Magenta
Life					L200
Essential		E20	E40	E100	E200
Discretionary	D6	D20	D40	D100	D200
Comfort	C6	C20	C40	C100	C200
	1–6	*7–20*	*21–40*	*41–100*	*101–200*
	Team Size				

Agile Process Overview

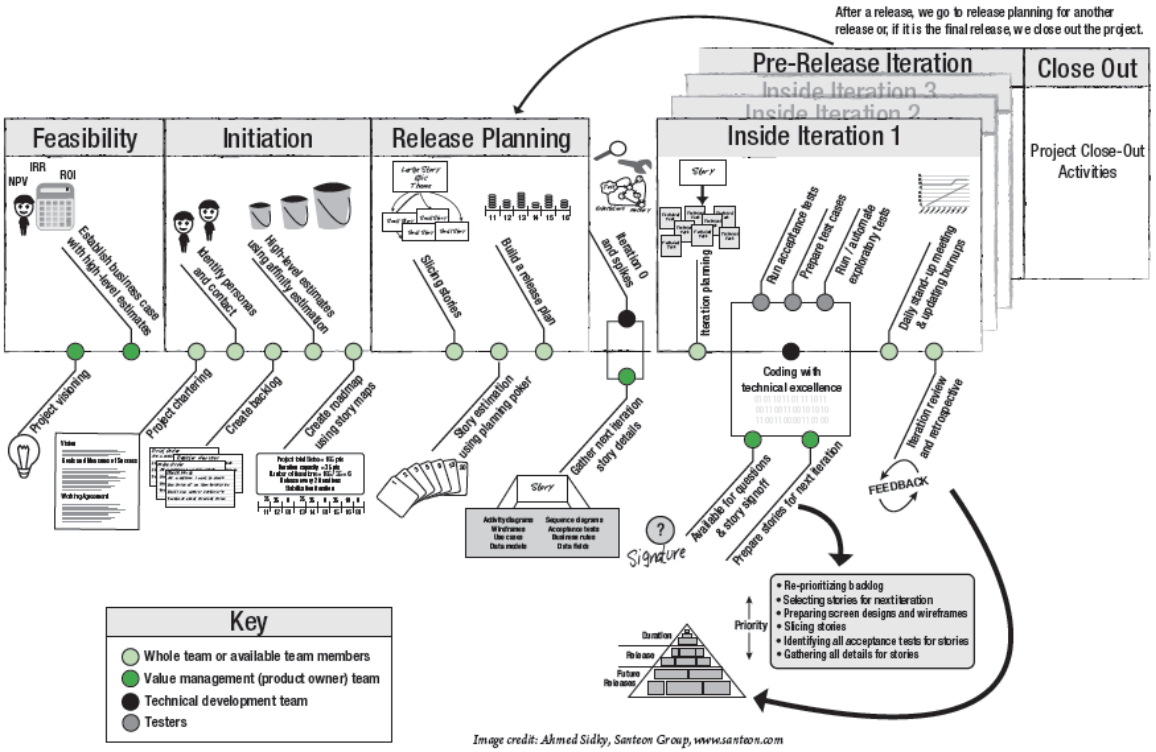


Image credit: Ahmed Sidky, Santeon Group, www.santeon.com

ASTANA IT UNIVERSITY

Agile Leadership

The concept of valuing people over processes impacts how leaders organize and motivate team members.

An effective leader will discover why team members want to do things, what motivates them, and align project tasks and goals accordingly.

Aligning project and personal objectives makes a more productive team.

Management Versus Leadership

Management: Mechanical focus concerned with tasks, control, and speed

Leadership: Humanistic focus on people and purpose, concerned with empowerment and effectiveness

ASTANA IT UNIVERSITY

Management Versus Leadership

4 primary duties:

1. Shield the team from interruptions: Protect the team from unnecessary diversions

2. Remove impediments to progress: Clear obstacles from the team's path that cause delay and waste

3. Communicate (and re-communicate) the project vision: Make project goals clear so team can align decisions with, and work toward, the common objective

4. Carry food and water: Provide the essential resources a team needs to keep them nourished and productive

ASTANA IT UNIVERSITY

12 Principles for Leading Agile Projects

1. Learn the team members' needs.
2. Learn the project's requirements.
3. Act for the simultaneous welfare of the team and the project.
4. Create an environment of functional accountability.
5. Have a vision of the completed project.
6. Use the project vision to drive your own behavior.

ASTANA IT UNIVERSITY

12 Principles for Leading Agile Projects (Continued)

7. Serve as the central figure in successful project team development.

8. Recognize team conflict as a positive step.

9. Manage with an eye toward ethics.

10. Remember that ethics is not an afterthought, but an integral part of our thinking.

11. Take time to reflect on the project.

12. Develop the trick of thinking backwards.

ASTANA IT UNIVERSITY

Agile Leadership Practices

Model desired behavior: Highest-ranked values desired in leaders are honesty, forward-looking, competent, and inspiring

Communicate the project vision: Frequently communicate project goals and objectives to ensure all stakeholders are aligned with vision

Enable others to act: Enable team members to feel confident in making decisions by fostering a collaborative environment

Be willing to challenge the status quo: Search for ways to change, grow, and improve

Leadership Tasks

Practice transparency through visualization: Be open and honest about progress, achievements, issues, and setbacks

Create a safe environment for experimentation: Team members are more likely to try new things if they know they won't be criticized

Experiment with new techniques and processes: Only way to find out if a new process works is to try it

Share knowledge through collaboration: Work with people to learn what they do and how they solve problems

Encourage emergent leadership via a safe environment: Take initiative and try a new approach after gaining team approval

ASTANA IT UNIVERSITY

References

Griffths M. PMI-ACP Exam Prep : A Course in a Book for Passing the PMI Agile Certified Practitioner (PMI-ACP) Exam (Updated Second Edition) Updated Second Edition (2018). RMC Publications, Inc.